

F-adopt-vs-rewrite

F — Adopt vs Refactor vs Rewrite vs Net-New decision table

Revision: 1 Last modified: 2026-06-10T00:00:00Z

Per-module disposition of the reference BobaLink implementation (44 .ts files under docs/research/browser_extension/Browser Torrent Extension Guide/src/src/) for the Boba browser-extension build. Decisions are evidence-based: every defect is cited to the reference file (and verified by direct read where the extraction's claim was ambiguous).

Decision vocabulary - **ADOPT** — reuse mostly as-is (rename/port only; no logic change). §11.4.74 reuse. - **REFACTOR** — reuse the structure, fix bounded defects (1–N targeted edits). - **REWRITE** — defects too deep / design wrong; rebuild from the contract, may salvage shape. - **NET-NEW** — does not exist in the reference; must be authored.

Anti-bluff (§11.4): a module is **ADOPTed** only when reading the source shows it is genuinely sound. Defective code is never **ADOPTed** just because it exists.

Reference path prefix (note the spaces), abbreviated <REF>/ below:
docs/research/browser_extension/Browser Torrent Extension Guide/
src/src/

1. Parser layer

Module/ File	Quality assessment (defect cited or “sound”)	Decision	Required fixes	Risk
parser/ bencode.ts	Sound. Zero-dep Uint8Array bencode; dict keys sorted lexicographically (bencode.ts:139-151) — load-bearing for stable infohash; rejects trailing data, unterminated ints/ strings/lists/dicts; bounds-checks byte- strings; binary-vs-utf-8 mode; sha1 via Web Crypto. Strongly unit- tested (bencode.test.ts).	ADOPT	Remove dead-assert in test (bencode.test.ts:156). Add property/fuzz + deep-recursion-limit (chaos) tests per §11.4.85. No source change.	Low
parser/ magnet.ts	Sound. Repeated-key query parsing (multiple tr/xt preserved), hex+base32 infohash, base32ToHex RFC4648, full param extraction (xt/dn/tr/ws/xl/xs/kt/as/ mt), recurse on base32. Fixture-driven tests.	ADOPT	Wire the unused 7- case invalidMagnets negative fixture. Consider BitTorrent v2 (btmh/multihash) — out of scope unless required.	Low
parser/ torrent- file.ts	Sound but DEAD CODE. parseTorrentFile/ computeInfohash correct (infohash = sha1(encode(infoDict))), BT v1 canonical) and tested — but never invoked by scanners/ adapter/background (sends upload raw File). Re-encode reorders non-sorted	REFACTOR	Source is sound → keep. The fix is wiring: invoke computeInfohash in the scanner/adapter to derive stable ids + local dedup against Boba. §11.4.124 (don’t delete unwired code — wire it).	Med (wiring touches scanner + adapter contracts)

Module/ File	Quality assessment (defect cited or “sound”)	Decision	Required fixes	Risk
	info keys (hash-divergence caveat, torrent-file.ts:46-48).			

2. Scanner layer

Module/File	Quality assessment	Decision	Required fixes	Risk
scanner/base.ts	One real defect. Abstract base + shadow-DOM-aware querySelectorAllDeep + hidden-element skip + re-entrancy guard are sound. Defect (verified by read): hashString returns Math.abs(hash).toString(36) + Date.now().toString(36) (base.ts:272) — time-salts the id despite the doc-comment “stable ID”, so the same torrent gets a new id every scan; cross-scan/cross-tab dedup is unreliable.	REFACTOR	Drop the Date.now() salt; derive id from infohash (now available once torrent-file.ts is wired) so ids are stable. One-method change.	Med (corrected dependency)
scanner/link-scanner.ts	Sound. Site-specific pass (resolve selectors by domain) + generic a[href] fallback, dedup by normalized href, relative→absolute resolution, sameOrigin flag.	ADOPT	Reconcile the selector source (see site-db row).	Low
scanner/text-scanner.ts	Sound. TreeWalker(SHOW_TEXT) with skip-list + min-length filter, findMagnetUri dedup. Catches forum-pasted magnets.	ADOPT	None.	Low
scanner/orchestrator.ts	Sound. Coordinates scanners + MutationObserver, per-site debounce, cooperative	ADOPT	Depends on the base.ts stable-id fix	Med (corrected)

Module/File	Quality assessment	Decision	Required fixes	Risk
scanner/site-db.ts	yieldToBrowser() between scanners, relevant-mutation filter, emits scan events, dedups by id into a Map. Initial-state-only tested (orchestrator never actually scans in the reference suite). Defect: two divergent selector tables. SITES (15 rich entries, used ONLY for per-site debounce) overlaps and disagrees with constants.SITE_SELECTORS (~21 flat entries, used by LinkScanner). The private trackers Boba actually supports (rutracker/iptorrents/kinozal/nmclub) are split across both.	REFACTOR	for reliable mergeResults dedup. Add real scan + mutation-rescan + debounce tests (the reference does not exercise scanning). Merge into ONE authoritative site registry (single source of truth) covering Boba's real trackers (rutracker, kinozal, nmclub, rutor, IPTorrents). Reconcile against the SQL site_selectors seed (30 rows) too.	Med (recon constants site-d SQL s

3. API layer

Module/File	Quality assessment	Decision	Required fixes
api/client.ts	Sound HTTP client, WRONG TARGET. TokenBucket rate-limit, AbortController timeout, retryWithBackoff, 429→RateLimitError, content-type	REFACTOR	Keep the transp (rate-limit/retry timeout/error mapping). Re-p base URL/

Module/File	Quality assessment	Decision	Required fixes
	branching — all sound. But it speaks ONLY raw qBittorrent WebUI v2 (/api/v2/...) and the default base assumes qBittorrent on :8080. Nothing reaches Boba's real services (7186 proxy / 7187 merge / 7189 jackett). set-cookie SID parse (client.ts:128-166) is browser-fragile.		endpoints to Boba's real ports and routes; add a Boba endpoint path separate from the raw qBt path set. Harden SID handling.
api/auth.ts	Mostly sound, one crypto-coupling defect. Four auth methods + failure backoff + refreshIfNeeded are sound. createCredentialsFromConfig(config, passphrase) (auth.ts:170) decrypts via shared/crypto — its correctness is hostage to the broken passphrase provenance (see crypto rows). Reads client["authCookie"] private field (auth.ts:247) — fragile TS escape hatch.	REFACTOR	Re-wire to the real key source (NEW below). Replace the private-field read with a public accessor on the client. Decide which auth method Boba actually uses (delegate to proxy/jackett vs raw q login).
api/qbittorrent.ts	Sound. Domain wrapper: sendTorrent/sendTorrents (250ms spacing), buildAddOptions maps ServerConfig→add params, never rethrows (normalizes to getUserMessage()).	ADOPT	Inherits the re-targeted client. None of its own logic is defective.
api/queue.ts	Defect: never authenticates (verified by read). Enqueue/dequeue/FIFO-eviction/persist sound. But processQueue (queue.ts:170) builds new BobaAPIClient(config.url,...) + adapter and runs the send loop with no login()/authenticate() call — retries against an authed server 403/fail-loop forever.	REFACTOR	Inject auth into processQueue (build+authenticate the client, or add an authenticate client/adapter). Add retry/attempt backoff tests (reference only) asserts structure
api/health.ts	Sound, wrong port list. Health thresholds, caching, testConnection, autoDiscover. Defect: autoDiscover probes [8080, 7187, 7189]	REFACTOR	Change probe port set to Boba's real map (7186/7187/7189)

Module/File	Quality assessment	Decision	Required fixes
	(health.ts:201) — 8080 is wrong for Boba (qBt WebUI is proxied on 7186).		

4. Content layer

Module/File	Quality assessment	Decision	Required fixes	Risk
content/ index.ts	Sound. Wires orchestrator→background (scan-result), message handler (scan-now/get-detected/toggle-selection/get-scan-status). Defect (contract): content get-detected returns {torrents:[...]} while background get-detected returns {data:{result}} — divergent shapes the popup juggles.	REFACTOR	Unify the get-detected response contract across content + background + popup.	Med
content/ scanner.ts	Sound. Thin content-side orchestrator glue (90 lines).	ADOPT	None.	Low
content/ highlight.ts	Sound. Three highlight styles (badge/border/glow), event-bus driven.	ADOPT	None (presentation).	Low
content/ styles.css	Presentation only, .boba-link-* + !important.	ADOPT	Rebrand class prefix if extension is renamed.	Low

5. Entrypoints (background / popup / options)

Module/ File	Quality assessment	Decision	Required fixes	Risk
		REFACTOR		

Module/ File	Quality assessment	Decision	Required fixes	Risk
background/ index.ts	Sound hub, ONE critical defect. Lifecycle, message router, context menus, commands, alarms (keepalive ~20s / health 5min), badge/ notifications — all sound. Critical (verified index.ts:409): sendTorrents decrypts creds with empty passphrase "" — can only ever read ""-encrypted bundles, and is mutually incompatible with the options page's "bobalink-extension" encryption ⇒ the whole credential path is broken end-to-end. open-dashboard fallback hardcodes :8080. Has zero unit coverage in the reference.		Replace the "" decrypt call-site with the real key source. Fix :8080 fallback. Add the missing background unit/integration coverage.	HIGH (credential correctness + the unrouted message types decision)

Module/ File	Quality assessment	Decision	Required fixes	Risk
popup/ popup.ts + popup/ index.html + popup/ styles.css	Sound. Per-tab list, select/ deselect, send, connection- status dot, empty-state scan. Excluded from reference coverage.	ADOPT	Adapt to the unified get- detected contract. Add real UI/ UX behavior tests (reference e2e asserts presence only).	Low-Med
options/ options.ts + index.html + styles.css	CRITICAL defect (verified options.ts:327). Form↔config bridge, server modal, test- connection, settings toggles all sound. But every secret is AES-encrypted with the literal fixed passphrase "bo balink- extension" — trivially reversible by anyone with the source (§11.4.10 leak). Auto- discover checkboxes are display-only (real ports hardcoded elsewhere).	REFACTOR	Remove the fixed passphrase; route encryption through the real key source (NET-NEW). Wire the auto- discover checkboxes to actual config.	HIGH (§11.4.10 credential leak is a release blocker)

6. Shared libs

Module/File	Quality assessment	Decision	Required fixes	Ri
shared/ constants.ts	Sound values, WRONG PORTS + duplication. Regexes, timeouts, retry/rate config, badge colors, encryption params all sound. Defects: DEFAULT_PORTS.QBITTORRENT=8080 / DEFAULT_URLS.QBITTORRENT wrong (should be 7186); SITE_SELECTORS duplicates site-db.SITES; RATE_LIMIT.MAX_REQUESTS=10 disagrees with DB seed behavior.rate_limit_requests=30. SOUND PRIMITIVE — extraction’s “REWRITE” is too broad (DISAGREE, see §8). Verified by read: PBKDF2 100k iters / AES-256-GCM / fresh 16-byte salt + 12-byte IV per op / 128-bit GCM tag / proper StorageError wrapping / rejects empty plaintext+passphrase. The crypto <i>module itself is correct and reusable</i>. The defect lives entirely in the callers’ passphrase (options.ts:327, background:409), not here.	REFACTOR	Fix qBt port 8080→7186 across host_permissions/CSP/URLs/PORTS/ auto-discover. Collapse the duplicate selector table. Pick canonical rate-limit.	Mo fix cu co ma SQ dis
shared/ crypto.ts	Sound. Typed error taxonomy (BobaLinkError + 8 subclasses), getUserMessage, normalizeError, isBobaLinkError.	ADOPT (module)	None to the module. The fix is the NET-NEW key source feeding it. Rewrite the test to import this real module (reference test tests a copy).	Lo (m Hi ke — NE
shared/ errors.ts	Sound. Typed EventEmitter + 13-event EventMap, per-listener try/catch isolation, unsub returns. Thoroughly tested.	ADOPT	Add direct error-class unit tests.	Lo
shared/ events.ts	Sound. Structured leveled logging, timed, createLogger, debug-gating.	ADOPT	None.	Lo
shared/ logger.ts	Sound. chrome.storage.local wrapper, NamespacedStorage,	ADOPT	Decide one persistence	Mo sto

Module/File	Quality assessment	Decision	Required fixes	Risk
	change listener, bobalink_-prefixed clear. Note: only the JSON-blob model is implemented here — the parallel SQL-schema (sql.js) model has NO wrapper file.		source of truth (JSON-blob vs sql.js); the sql.js wrapper is NET-NEW if chosen. Add direct storage unit tests (always mocked away in the reference).	ar de
shared/ utils.ts	Sound. debounce/throttle, retryWithBackoff, TokenBucket, processInChunks, formatBytes, url helpers. Note: escapeHtml is DOM-dependent (not service-worker safe); processInChunks defined but unused.	ADOPT	Provide a SW-safe escapeHtml variant for background context.	Lo

7. Types

Module/ File	Quality assessment	Decision	Required fixes	Risk
types/ torrent.ts	Sound. MagnetInfo/ TorrentFile/ ParsedTorrent/ DetectedTorrent/ SendResult/ PageScanResult.	ADOPT	None.	Low
types/ config.ts	Sound, with duplication. ServerConfig/ ExtensionConfig/ DEFAULT_CONFIG. Defect: AuthMethod duplicated in types/api.ts; ServerConfig (TS) diverges from server_config (SQL) field naming.	REFACTOR	De-duplicate AuthMethod (single canonical). Reconcile TS ServerConfig ↔ SQL server_config. Fix default ports (auto-discovery [7187, 7189, 8080] → Boba map).	Low

Module/ File	Quality assessment	Decision	Required fixes	Risk
types/ api.ts	Sound, declares unused contracts. qBt/auth/health/queue/message types. BobaServerInfo/ BobaSearchResponse declared but nothing consumes them (no Boba-API client). MessageType declares 7 values with no handler branch.	REFACTOR	De-dup AuthMethod. Decide which declared-but-unrouted message types to implement vs drop. The Boba-API consumer that uses BobaSearchResponse is NET-NEW.	Low (types) / High (the consumer is net-new)

8. SQL schema

Module/File	Quality assessment	Decision	Required fixes	Risk
sql/schemas.sql (= migrations/ 001_initial.sql)	Sound schema, wrong default + unresolved dual-storage. 9 tables, ~25 indexes, partial indexes, encrypted-credential columns (api_key_encrypted/password_encrypted), 30 selector seeds, 25 config seeds — well-formed SQLite 3.38+/sql.js. Defects: seeds qbittorrent_port DEFAULT 8080 + server.base_url=http://localhost:8080; and the runtime has NO sql.js wrapper (only the JSON-blob storage path is implemented) so it is	REFACTOR (if SQL chosen) / drop (if JSON-blob chosen)	First decide the single persistence model. If SQL: fix port seeds 8080→7186, reconcile site-selector seed with the merged site registry, author the sql.js wrapper (NET-NEW). If JSON-blob: this	Med (architecture decision gates everything)

Module/File	Quality assessment	Decision	Required fixes	Risk
	currently unused-by-design.		schema is not adopted.	

9. Build / test configs

Module/File	Quality assessment	Decision	Required fixes
wxt.config.ts	Sound MV3 manifest gen, wrong endpoints. WXT manifest, 6 permissions, CSP, commands, auto-icons. Defects: host_permissions + CSP connect-src include :8080 (should be :7186); icon PNGs referenced but absent.	REFACTOR	Fix 8080→7186 in host_permission + CSP. Confirm Firefox MV3 module-SW support.
tsconfig.json	Sound. Very strict (all strict flags + noUncheckedIndexedAccess + exactOptionalPropertyTypes).	ADOPT	None.
package.json	Sound, drift. Deps/scripts fine. Defects: HTTPS repo URL (Boba mandates SSH); release.yml references non-existent build:chrome script.	REFACTOR	SSH remote; rebrand name; align scripts; drop CI-only assumptions.
jest.config.ts	Defect: 80% threshold + UI/entrypoints excluded from coverage. collectCoverageFrom excludes popup/options/content/background/**/index.ts — UI/entrypoint logic has ZERO coverage enforcement. Conflicts with §11.4.27.	REWRITE (or replace w/ Vitest)	Decide runner (Vitest recommended — matches Boba frontend/ + the crypto test). Bring UI/entrypoints INTO coverage; raise toward 100%.
playwright.config.ts	Defect: non-functional e2e config. References missing global-setup.ts/global-teardown.ts; dead webServer.url :3000; Firefox	REWRITE	Author global-setup/teardown (build dist, launch persistent

Module/File	Quality assessment	Decision	Required fixes
	project can't load extension via Chromium args; reporter:github+forbidOnly:CI assume GitHub Actions.		Chromium with --load-extension, capture real extension id, rewrite chrome-extension:// <id>/...). Remove GH- Actions assumptions (Boba manual CI).
.eslintrc.json / .prettierrc	Sound. Type-aware lint (no-explicit-any/no-floating-promises errors), Prettier 100-col.	ADOPT	Align with Boba project lint config.
_locales/en/ messages.json	Sound. 28 i18n keys.	ADOPT	Rebrand strings if renamed.
.github/workflows/ ci.yml, release.yml	FORBIDDEN. Boba Hard-Stop: no .github/workflows/*.yml ever.	drop (do NOT port)	Extract only the <i>commands</i> (lint/ compile/test/ build/zip/sha256 into a manual ci.sh-style script. Never copy the YAML.

10. Tests

Module/File	Quality assessment	Decision	Required fixes	Risk
bencode.test.ts	Sound; dead-assert line 156.	ADOPT	Remove dead assert; add fuzz/chaos.	Low
magnet.test.ts	Sound, fixture-driven; invalidMagnets (7) unused.	ADOPT	Wire negative fixtures.	Low
		ADOPT	None.	Low

Module/File	Quality assessment	Decision	Required fixes	Risk
torrent-file.test.ts	Sound (infohash determinism, single/multi parse).			
api-client.test.ts	Partial BLUFF. Auth block ends expect(true).toBe(true) (no observable check); 429 test asserts ServerError despite name saying RateLimitError.	REFACTOR	Replace the bluff Auth assertions with real cookie-header propagation checks; fix the 429 assertion.	Med
queue.test.ts	Partial BLUFF. Auto-processing block ends expect(true).toBe(true); processQueue only asserts <i>structure</i> , not success/failure.	REFACTOR	Assert real retry/attempts/outcomes; remove no-ops.	Med
scanner.test.ts	Weak. Event-emitter block sound; but orchestrator block asserts initial-state only (never scans); DOM block tests raw querySelectorAll, NOT the production scanner.	REFACTOR	Drive the real orchestrator over real DOM; assert detected torrents + mutation rescans.	Med
crypto.test.ts	BLUFF (§11.4.1). 44 thorough tests — but against an inline re-implemented copy , never src/shared/crypto.ts. Cannot fail against a no-op stub of the real module. Also imports vitest while the other 6 files use Jest (mixed runners).	REWRITE	Re-point every test at the production shared/crypto module; standardize runner.	Med (it's the most-thorough file but proves nothing as-is)
chrome-mock.ts / setup.ts	Mock infra; happy-path only (no error injection), no cookies/scripting/	REFACTOR	Add error-injecting variant for chaos/stress;	Low

Module/File	Quality assessment	Decision	Required fixes	Risk
content.spec.ts / options.spec.ts / popup.spec.ts (e2e)	sync; setup.ts:14 wrong path (type-only). NON-FUNCTIONAL. Hardcoded chrome- extension://test-id/...; no fixture loads the extension or resolves the real id; content spec asserts raw DOM, not the injected content script. Asserts presence/ visibility, never user outcomes.	REWRITE	fix the stale path. Real extension- loading fixture + user- observable assertions (torrent rendered, server persisted, send→appears in qBittorrent).	High (load- bearing for §11.4 anti- bluff e2e)

11. NET-NEW (does not exist in the reference — must be authored)

Item	Why net-new	Risk
Real key source for credential encryption	crypto.ts is a sound primitive but has NO real passphrase provenance — callers use a fixed string + an empty string. Need a master- passphrase prompt, OS-keychain, OR delegation to boba- jacket/proxy (which already own encrypted creds at / config/boba.db). §11.4.10 blocker.	HIGH
Real Boba service integration (7186/7187/7189)	Reference speaks ONLY raw qBittorrent v2; BobaServerInfo/	HIGH

Item	Why net-new	Risk
	BobaSearchResponse types are declared but nothing consumes them. The merge-search API (7187), tracker-cookie download path (7186 proxy), and boba-jackett (7189) clients do not exist. Verified: the reference has NO chrome.tabGroups usage. The only “batching” is intra-tab yieldToBrowser() cooperative yielding + per-tab tabTorrents map + an unused processInChunks. Tab-group batch behavior is fresh design.	
chrome.tabGroups tab-group batching		Med
Icon PNG rasterization (16/32/48/128)	Verified: src/assets/ contains icon.svg only — zero .png files. Code/manifest reference rasterized PNGs that must be produced at build (WXT auto-icons from the SVG).	Low
sql.js storage wrapper (only if SQL persistence chosen)	The 9-table schema exists but NO runtime wrapper implements it; only the JSON-blob storage.ts path is live.	Med
Playwright global-setup.ts / global-teardown.ts	Referenced by config, absent from tree. Must build dist, launch persistent	Med

Item	Why net-new	Risk
	Chromium --load-extension, capture real extension id.	
Missing test TYPES (§11.4.27/§11.4.85): integration (real cross-module → live 7186/7187), functional e2e, automation/HelixQA, security/pen (credential-leak, XSS via magnet dn, CSP, permission-scope), DDoS/load, scaling, chaos (storage corruption / network fault / SW termination), stress, performance, benchmark, UI/UX, Challenges	Reference covers only unit + e2e (and those have bluffs). 13 of 15 mandated types absent.	HIGH (volume)

Reuse estimate

Counting the **44 .ts source files** plus the SQL schema and the build/test configs:

Disposition	Approx. count	Examples
ADOPT (reuse ~as-is)	~17	bencode, magnet, link/text-scanner, content/scanner+highlight, qbittorrent adapter, crypto module, errors, events, logger, storage, utils, types/torrent, eslint/prettier/tsconfig, i18n
	~18	

Disposition	Approx. count	Examples
REFACTOR (bounded fixes)		torrent-file (wire), base (un-salt id), site-db (merge tables), client (re-target), auth, queue (auth), health (ports), content/index (contract), background, options, popup, constants (ports), types/config, types/api, wxt.config, package.json, chrome-mock, sql schema
REWRITE	~7	jest.config (runner + coverage), playwright.config, crypto.test, api-client.test (bluff), queue.test (bluff), scanner.test, all 3 e2e specs
NET-NEW	7+ areas	key source, Boba-service clients, tab-groups, icon PNGs, sql.js wrapper, e2e global setup, 13 missing test types
DROP	2	both .github/workflows/*.yaml (constitution hard-stop)

Headline: roughly 80% of the reference's logic/structure is reusable (ADOPT + REFACTOR $\approx$ 35 of ~44 source-bearing files — the parsers, scanners, shared libs, types, and the API transport are sound), **~20% must be rewritten** (the test suite's bluffs/non-functional e2e + the runner/coverage configs), **and the highest-value work is NET-NEW integration** (real Boba endpoints + a real key source) that the reference never attempted. The reference is a strong *skeleton with*

sound primitives whose two structural gaps — credential key-management and Boba-service integration — are exactly the parts it left as placeholders.

Top 5 highest-risk rewrites

1. **Real key source for credential encryption (NET-NEW).** The sound crypto primitive is currently fed a fixed ("boba-link-extension", options.ts:327) and an empty ("", background:409) passphrase that are even mutually incompatible — the credential path is broken end-to-end and is a §11.4.10 leak (release blocker). Touches options + background + auth.
2. **Boba-service integration / api/client.ts re-target (REFACTOR→effectively rewrite of the endpoint layer).** The client speaks only raw qBittorrent v2 on :8080; nothing reaches 7186/7187/7189. Re-targeting + adding the Boba-API consumer (BobaSearchResponse) is the load-bearing integration, and the SID/cookie + CORS/host-permission story is fragile.
3. **Functional e2e suite (REWRITE all 3 specs + author global-setup/teardown).** Every reference e2e is non-functional (hardcoded chrome-extension://test-id, no extension load, presence-only asserts). §11.4 anti-bluff demands real extension load + user-observable outcomes — this is from-scratch.
4. **crypto.test.ts (REWRITE).** The single most thorough test file (44 cases) is a §11.4.1 bluff — it tests an inline copy, never the production module, so it passes against a no-op stub. Must be re-pointed at shared/crypto and the runner standardized (mixed Jest/Vitest).
5. **Test runner + coverage config (jest.config REWRITE).** 80% threshold with UI/entrypoints excluded conflicts with §11.4.27; the runner decision (Vitest vs Jest) is foundational and gates every other test file's port. (Close behind: queue.ts auth-gap fix + the base.ts time-salted-id fix, both correctness-critical for offline retries and dedup.)

Disagreements with the extraction's defect claims (re-verified by reading source)

1. **crypto.ts → the extraction tags it “ REWRITE”; I DISAGREE — the MODULE is ADOPT, the KEY SOURCE is NET-NEW.** Direct read of shared/crypto.ts (lines 42-228) shows the encrypt/decrypt

machinery is genuinely sound: PBKDF2 100k iterations, AES-256-GCM, a fresh 16-byte random salt + 12-byte IV per operation, a 128-bit GCM auth tag, and correct `StorageError` wrapping that rejects empty plaintext/passphrase. **No defect exists inside this file.** The defects the extraction cites — fixed passphrase `"bobalink-extension"` (verified `options.ts:327`) and empty passphrase `""` (verified `background/index.ts:409`) — live entirely in the *callers*, not in the `crypto` module. So the correct disposition is: **ADOPT the crypto primitive unchanged; author the real key source as NET-NEW; REFACTOR the two call-sites.** Rewriting the `crypto` module itself would be discarding sound, reusable work (§11.4.74). The extraction's "`crypto ...` → REWRITE" headline conflates the module with its misuse.

2. **Minor: the extraction calls `base.ts` id generation a generic “unstable / time-salted” concern; verified it is a one-line REFACTOR, not a rewrite.** `base.ts:272` literally appends `Date.now().toString(36)` to the hash; removing that suffix and sourcing the id from the (now-wireable) `infohash` is a bounded fix to a single private method — the rest of `base.ts` (shadow-DOM traversal, hidden-element filtering, re-entrancy guard) is sound and ADOPT-grade.

All other extraction defect claims I sampled were verified accurate: the `queue` never authenticates (`queue.ts:170` builds+runs with no login), the `.torrent` parser is genuinely never invoked, the two selector tables genuinely diverge, the icon PNGs are genuinely absent (only `icon.svg`), and the two passphrase call-sites are genuinely incompatible.